Chapter 8

Gradient Flows

In this chapter we present one of the most spectacular application of optimal transport and Wasserstein distances to PDEs. We will see that several evolution PDEs can be interpreted as a steepest descent movement in the space $\mathbb{W}_2$. As we already developed in the previous chapters many of the technical tools that we need, a large part of this chapter will be devoted to an informal description of the general framework.

Hence, no need to let the reader wait any more, let us immediately enter into the subject!

8.1 Gradient flows in $\mathbb{R}^d$ and in metric spaces

First of all, let us present what a gradient flow is in the simplest situation. Suppose you have a function $F : \mathbb{R}^d \to \mathbb{R}$ and a point $x_0 \in \mathbb{R}^d$. A gradient flow is an evolution stemming from x_0 and always moving in the direction where F decreases the most, thus "gradually minimizing" F, starting from x_0. More precisely, it is just the solution of the Cauchy problem

$$\begin{cases} x'(t) = -\nabla F(x(t)) & \text{for } t > 0, \\ x(0) = x_0. \end{cases}$$

This is a standard Cauchy problem which has a unique solution if ∇F is Lipschitz continuous, i.e. if $F \in C^{1,1}$. We will see that existence and uniqueness can also hold without this strong assumption, thanks to the variational structure of the equation.

A first interesting property is the following, concerning uniqueness and estimates.

Proposition 8.1. *Suppose that F is convex and let x_1 and x_2 be two solutions of $x'(t) = -\nabla F(x(t))$ (if F is not differentiable we can consider $x'(t) \in \partial F(x(t))$). Then we have $|x_1(t) - x_2(t)| \le |x_1(0) - x_2(0)|$ for every t. In particular this gives uniqueness of the solution of the Cauchy problem.*

Proof. Let us consider $g(t) = \frac{1}{2}|x_1(t) - x_2(t)|^2$ and differentiate it. We have

$$g'(t) = (x_1(t) - x_2(t)) \cdot (x_1'(t) - x_2'(t)) = -(x_1(t) - x_2(t)) \cdot (\nabla F(x_1(t)) - \nabla F(x_2(t))).$$

Here we use the basic property of gradient of convex functions, i.e. that for every x_1 and x_2 we have

$$(x_1 - x_2) \cdot (\nabla F(x_1) - \nabla F(x_2)) \geq 0.$$

More generally, it is also true that for every x_1, x_2 and every $p_1 \in \partial F(x_1)$, $p_2 \in \partial F(x_2)$, we have

$$(x_1 - x_2) \cdot (p_1 - p_2) \geq 0.$$

From these considerations, we obtain $g'(t) \leq 0$ and $g(t) \leq g(0)$. This gives the first part of the claim.

Then, if we take two different solutions of the same Cauchy problem, we have $x_1(0) = x_2(0)$, and this implies $x_1(t) = x_2(t)$ for any $t > 0$. □ □

Remark 8.2. From the same proof, one can also deduce uniqueness and stability estimates in the case where F is only semi-convex. We recall that semi-convex means that it is λ-convex for some $\lambda \in \mathbb{R}$ i.e. $x \mapsto F(x) - \frac{\lambda}{2}|x|^2$ is convex. Indeed, in this case we obtain $|x_1(t) - x_2(t)| \leq |x_1(0) - x_2(0)|e^{-\lambda t}$, which also proves, if $\lambda > 0$, exponential convergence to the unique minimizer of F. The key point is that, if F is λ-convex it is easy, by applying the monotonicity inequalities above to $x \mapsto F(x) - \frac{\lambda}{2}|x|^2$, to get

$$(x_1 - x_2) \cdot (\nabla F(x_1) - \nabla F(x_2)) \geq \lambda |x_1(t) - x_2(t)|^2.$$

This implies $g'(t) \leq -2\lambda g(t)$ and allows to conclude, by Gronwall's lemma, $g(t) \leq g(0)e^{-2\lambda t}$. For the exponential convergence, if $\lambda > 0$ then F is coercive and admits a minimizer, which is unique by strict convexity. Let us call it $\bar{x}$. Take a solution $x(t)$ and compare it to the constant curve $\bar{x}$, which is a solution since $0 \in \partial F(\bar{x})$. Then we get $|x_1(t) - \bar{x}| \leq e^{-\lambda t}|x_1(0) - \bar{x}|$.

Another interesting feature of those particular Cauchy problems which are gradient flows is their discretization in time. Actually, one can fix a small time step parameter $\tau > 0$ and look for a sequence of points $(x_k^\tau)_k$ defined through

$$x_{k+1}^\tau \in \operatorname{argmin}_x F(x) + \frac{|x - x_k^\tau|^2}{2\tau}.$$

We can forget now the convexity assumptions on F, which are not necessary for this part of the analysis. Indeed, very mild assumptions on F (l.s.c. and some lower bounds, for instance $F(x) \geq C_1 - C_2|x|^2$) are sufficient to guarantee that these problems admit a solution for small τ. The case where F is λ-convex is covered by these assumptions, and also provides uniqueness of the minimizers. This is evident if $\lambda > 0$ since we have strict convexity for every τ, and if λ is negative the sum will be strictly convex for small τ.

We can interpret this sequence of points as the values of the curve $x(t)$ at times $t = 0, \tau, 2\tau, \ldots, k\tau, \ldots$. It happens that the optimality conditions of the recursive minimization exactly give a connection between these minimization problems and the equation, since we have

$$x_{k+1}^\tau \in \operatorname{argmin} F(x) + \frac{|x - x_k^\tau|^2}{2\tau} \Rightarrow \nabla F(x_{k+1}^\tau) + \frac{x_{k+1}^\tau - x_k^\tau}{\tau} = 0,$$

i.e.

$$\frac{x_{k+1}^\tau - x_k^\tau}{\tau} = -\nabla F(x_{k+1}^\tau).$$

This expression is exactly the discrete-time implicit Euler scheme for $x' = -\nabla F(x)$!

> *Box* 8.1. **– Memo** – Explicit and Implicit Euler schemes
>
> Given an ODE $x'(t) = \mathbf{v}(x(t))$ (that we take autonomous for simplicity), with given initial datum $x(0) = x_0$, Euler schemes are time-discretization where derivatives are replaced by finite differences. We fix a time step $\tau > 0$ and define a sequence x_k^τ. The explicit scheme is given by
>
> $$x_{k+1}^\tau = x_k^\tau + \tau \mathbf{v}(x_k^\tau), \quad x_0^\tau = x_0.$$
>
> The implicit scheme, on the contrary, is given by
>
> $$x_{k+1}^\tau = x_k^\tau + \tau \mathbf{v}(x_{k+1}^\tau), \quad x_0^\tau = x_0.$$
>
> This means that x_{k+1}^τ is selected as a solution of an equation involving x_k^τ, instead of being explicitly computable from x_k^τ. The explicit scheme is obviously easier to implement, but enjoys less stability and qualitative properties than the implicit one. Suppose for instance $\mathbf{v} = -\nabla F$: then the quantity $F(x(t))$ decreases in t in the continuous solution, which is also the case for the implicit scheme, but not for the explicit one (which represents the iteration of the gradient method for the minimization of F).
>
> Note that the same can be done for evolution PDEs, and that solving the Heat equation $\partial_t \rho = \Delta \rho_t$ by an explicit scheme is very dangerous: at every step, ρ_{k+1}^τ would have two degrees of regularity less than ρ_k^τ, since it is obtained through $\rho_{k+1}^\tau = \rho_k^\tau - \Delta \rho_k^\tau$.

It is possible to prove that, for $\tau \to 0$, the sequence we found, suitably interpolated, converges to the solution of the problem. It even suggests how to define solutions for functions F which are only l.s.c., with no gradient at all!

But a huge advantage of this discretized formulation is also that it can easily be adapted to metric spaces. Actually, if one has a metric space (X,d) and a l.s.c. function $F : X \to \mathbb{R} \cup \{+\infty\}$ bounded from below, one can define

$$x_{k+1}^\tau \in \operatorname{argmin}_x F(x) + \frac{d(x,x_k^\tau)^2}{2\tau} \tag{8.1}$$

and study the limit as $\tau \to 0$. Obviously the assumptions on F have to be adapted, since we need existence of the minimizers and hence a little bit of compactness. But if X is compact then everything works for F only l.s.c.

We can consider two different interpolations of the points x_k^τ, given respectively by:

$$x^\tau(t) = x_k^\tau, \qquad \widetilde{x}^\tau(t) = \omega_{x_{k-1}^\tau, x_k^\tau}\left(\frac{t-(k-1)\tau}{\tau}\right) \quad \text{for } t \in](k-1), \tau, k\tau],$$

where $\omega_{x,y}(s)$ denotes any constant speed geodesic connecting a point x to a point y, parametrized on the unit interval $[0,1]$. The interpolation $\widetilde{x}^\tau$ only makes sense in spaces where geodesics exist, obviously. It is in this case a continuous (locally Lipschitz) curve, which coincides with the piecewise constant interpolation x^τ at times $t = k\tau$.

De Giorgi, in [138], defined[1] the notion of *Minimizing Movements* (see also [7]):

Definition 8.3. A curve $x : [0,T] \to X$ is said to be a Minimizing Movement if there exists a sequence of time steps $\tau_j \to 0$ such that the the piecewise constant interpolations x^{τ_j}, built from a sequence of solutions of the iterated minimization scheme (8.1), uniformly converge to x on $[0,T]$.

Compactness results guaranteeing the existence of Minimizing Movements are derived from a simple property giving a Hölder behaviour for the curves x^τ: for every τ and every k, the optimality of x^τ_{k+1} provides

$$F(x^\tau_{k+1}) + \frac{d(x^\tau_{k+1}, x^\tau_k)^2}{2\tau} \le F(x^\tau_k), \tag{8.2}$$

which implies

$$d(x^\tau_{k+1}, x^\tau_k)^2 \le 2\tau \left(F(x^\tau_k) - F(x^\tau_{k+1})\right).$$

If $F(x_0)$ is finite and F is bounded from below, taking the sum over k, we have

$$\sum_{k=0}^{l} d(x^\tau_{k+1}, x^\tau_k)^2 \le 2\tau \left(F(x^\tau_0) - F(x^\tau_{l+1})\right) \le C\tau.$$

Given $t \in [0,T]$, denote by $\ell(t)$ the unique integer such that $t \in](\ell(t)-1)\tau, \ell(t)\tau]$. For $t < s$ apply the Cauchy-Schwartz inequality to the sum between the indices $\ell(t)$ and $\ell(s)$ (note that we have $|\ell(t) - \ell(s)| \le \frac{|t-s|}{\tau} + 1$)

$$\begin{aligned} d(x^\tau(t), x^\tau(s)) &\le \sum_{k=\ell(t)}^{\ell(s)-1} d(x^\tau_{k+1}, x^\tau_k) \le \left(\sum_{k=\ell(t)}^{\ell(s)-1} d(x^\tau_{k+1}, x^\tau_k)^2\right)^{\frac{1}{2}} |\ell(s) - \ell(t)|^{\frac{1}{2}} \\ &\le C\sqrt{\tau}\frac{(|t-s|+\tau)^{\frac{1}{2}}}{\sqrt{\tau}} \le C\left(|t-s|^{1/2} + \sqrt{\tau}\right). \end{aligned}$$

This means that the curves x^τ - if we forget for a while that they are actually discontinuous - are morally equi-Hölder continuous of exponent $1/2$. The situation is even clearer for $\widetilde{x}^\tau$. Indeed, we have

$$\tau \left(\frac{d(x^\tau_{k-1}, x^\tau_k)}{\tau}\right)^2 = \int_{(k-1)\tau}^{k\tau} |(\widetilde{x}^\tau)'|^2(t)\mathrm{d}t,$$

which implies, by summing up

$$\int_0^T |(\widetilde{x}^\tau)'|^2(t)\mathrm{d}t \le C.$$

[1] We avoid here the distinction between Minimizing Movements and Generalized Minimizing Movements, which is not crucial in our analysis.

This means that the curves $\widehat{x}^\tau$ are bounded in $H^1([0,T];X)$ (the space of absolutely continuous curves with square-integrable metric derivative), which also implies a $C^{0,\frac{1}{2}}$ bound by usual Sobolev embedding.

If the space X, the distance d, and the functional F are explicitly known, in some cases it is already possible to pass to the limit in the optimality conditions of each optimization problem in the time-discrete setting, and to characterize the limit curve $x(t)$. It will be possible to do so in the space of probability measures, which is the topic of this chapter, but not in many other cases. Indeed, without a little bit of (differential) structure on X, it is almost impossible. If we want to develop a general theory of gradient flows in metric spaces, we need to exploit finer tools, able to characterize, with the only help of metric quantities, the fact that a curve $x(t)$ is a gradient flow.

We present here an inequality which is satisfied by gradient flows in the smooth Euclidean case, and which can be used as a definition of gradient flow in the metric case, since all the quantities which are involved have their metric counterpart. Another inequality, called EVI, will be presented in the discussion section (8.4.1).

The first observation is the following: thanks to the Cauchy-Schwartz inequality, for every curve $x(t)$ we have

$$F(x(s))-F(x(t))=\int_s^t -\nabla F(x(r))\cdot x'(r)\,dr \le \int_s^t |\nabla F(x(r))||x'(r)|\,dr$$
$$\le \int_s^t \left(\frac{1}{2}|x'(r)|^2+\frac{1}{2}|\nabla F(x(r))|^2\right)dr.$$

Here, the first inequality is an equality if and only if $x'(r)$ and $\nabla F(x(r))$ are vectors with opposite directions for a.e. r, and the second is an equality if and only if their norms are equal. Hence, the condition, called EDE (*Energy Dissipation Equality*)

$$F(x(s))-F(x(t))=\int_s^t \left(\frac{1}{2}|x'(r)|^2+\frac{1}{2}|\nabla F(x(r))|^2\right)dr, \quad \text{for all } s<t$$

(or even the simple inequality $F(x(s))-F(x(t))\ge\int_s^t\left(\frac{1}{2}|x'(r)|^2+\frac{1}{2}|\nabla F(x(r))|^2\right)dr$) is equivalent to $x'=-\nabla F(x)$ a.e., and could be taken as a definition of gradient flow. This is what is done in a series of works by Ambrosio, Gigli and Savaré, and in particular in their book [17]. Developing this (huge) theory is not among the scopes of this book. The reader who is curious about this abstract approach but wants to find a synthesis of their work from the point of view of the author can have a look at [275]. The role of the different parts of the theory with respect to the possible applications is clarified as far as possible[2] (we also discuss in short some of these issues in Section 8.4.1).

8.2 Gradient flows in $\mathbb{W}_2$, derivation of the PDE

In this section we give a short and sketchy presentation of what can be done when we consider the gradient flow of a functional $F:\mathscr{P}(\Omega)\to\mathbb{R}\cup\{+\infty\}$. The functional will

[2] Unfortunately, some knowledge of French is required (even if not forbidden, English is unusual in the Bourbaki seminar, since "Nicolas Bourbaki a une préférence pour le français").